Federated Learning Approach for Obstructive Sleep Apnea Prediction

Bachelor thesis presentation

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Outline

Introduction

- Team
  - Manja Gorenc Novak, *Head of AI and Senior Data Scientist*
  - Rok Vinder, *Data Analyst and Machine Learning Engineer*
- Result d.o.o
  - Slovenian Company
  - $\sim$ 50 employees
  - Software development
  - Data and AI
  - Business solutions
- Context
  - eEarly
    - Health data
    - Obstructive sleep apnea

Project background

Problem statement

- Problem statement
  - Machine learning
  - Centralized
  - Private (health) data
  - Regulations

Project background

Stakeholders and their needs

- **Developers**
  - Understand code
  - Interpret results
  - Run the project
- *Patients*
  - **Data privacy**
- *Doctors*
  - *Improved diagnosis*
  - *Better tools*

Objective within the Bachelor Thesis

Proposed solution

- Proposed solution
  - Federated learning
  - ~~Data to the model (centralized)~~
  - Model to the data (decentralized)
  - Privacy, Regulations
  - Bandwidth

Objective within the Bachelor Thesis

Federated learning algorithm

Objective within the Bachelor Thesis

Scope

Table: Red color represents MVP, orange and blue represent broader or ultimate scope.

Importance	Requirement
Must have	Single-machine and non-containerized simulation of apnea predictions using federated learning (POC).
Should have	Single-machine and containerized (with Docker) −‖−
Could have	Multi-machine and containerized (with Docker) −‖−
Could have	Multi-machine and containerized (on Kubernetes) −‖−
Could have	Differential privacy mechanism implemented.

Research and/or development

Comparative study and technology decision

Flower

- beginner friendly
- prototype friendly
- good documentation
- excellent integration (!)

NVIDIA FLARE

- enterprise ready
- many built-in features
- NVIDIA ecosystem
- heavy for small projects

TensorFlow Federated

- widely used
- complex
- the least integration

Solution

Final solution

- Federated learning network
- Single-machine (non-containerized)
- Multi-machine (containerized)

Project Approach

Development method

Scrumban

- Daily meetings
- Agile mindset
- Backlog
- Typical kanban board

Project Approach

Work division and Collaboration

Work division

- MVP
  - Flower quickstart examples
  - Connecting with existing apnea prediction model
  - Preparing the data
- Additional features
  - Containerizing with Docker
  - Connecting multiple machines

Collaboaration

- Daily meetings
- Slack and Outlook

Output and Demo

Reflection

Achieved objectives

Table: Red color represents MVP, orange and blue represent broader or ultimate scope.

Importance	Requirement	Done
Must have	Single-machine and non-containerized simulation of apnea predictions using federated learning.	Yes
Should have	Single-machine and containerized (with Docker) –||–	Yes
Could have	Multi-machine and containerized (with Docker) –||–	Yes
Could have	Multi-machine and containerized (on Kubernetes) –||–	No
Could have	Differential privacy mechanism implemented.	No

Reflection

Added value and impact

- Privacy (not only patients)
- Better diagnosis tools (doctors)
- Regulations (developers)
- Less network usage (climate)
- Project opportunities (Result)
- Research (scientists)

Reflection

Improvements, and choice reflection

Improvements and weaknesses

- Evaluation metrics
- Separate networks
- Real TLS certification

Choice reflection

- Excellent
- No change

Thank you for your attention!

Questions and Answers